■■■ — ■■■■■ ■■ ■■■■■ ■■■ ■■■

> ■■■ ■■■■■ ■■■■ ■■ ■■■ ■■■■■.

---

1. ■■ ■■■

■■■■■ ■■

■■■■ ■■■ ■■■■■ ■■■·■■■ ■■■ ■ 2■ ■■ ■■■ ■■■■. ■■ ■■ ■ ■■■■ ■■ ■■■■■■ ■■■■ ■■■ ■■ ■ ■■ ■■■ ■■■.

■■	■■	■■ ■■
■■ ■■ ■■	■■ ■ ■■ ■■■ 30–60■ ■■	■■■ ■■ ■■ ■■
■■■·■■ ■■	■■ ■■■ ■■ ■■■ ■ ■	ROAS·■■■ ■■ ■■ ■■
■■ ■■ ■■■	■■■■ ■■ ■■■ ■■■■ ■■ ■■	■■ ■■ ■■, ■■■■■ ■■ ■■

■■ ■■ ■■

■ ■■ ■ ■■■·■■ ■■■ 1■■■ ■■■■. ■■ ■■■ ■■■■ ■■■■ ■■■■■■, ■■■ ■■■ ■■ ■■(ROAS, ■■■)■ ■■■■■ ■■■■ ■■■■ ■■ ■■. ■■ ■■ ■■■ ■■■■ ■■ ■■■ ■■■ ■■■■■ ■■ ■■■ ■■ ■■■ ■■■■.

---

2. ■■■ ■■■■ ■■■■■

```
[■■ mp3]
   ■
   ▼
[■■] faster-whisper (STT) + pyannote 3.1 (■■■■)
   ■  → canonical Transcript JSON
   ■
   ▼
[■■·■■] ■■ ■■■ → ■■■ → SQLite / PostgreSQL
   ■  meetings / utterances / chunks / action_item_sources
   ■
   ▼
[AI ■■] Gemini 2.5 Flash Lite  ■■■■■■■  Mock Extractor
   ■  structured JSON (■■■·■■·■■■■·■■■·■■■■)
   ■
```

```
      ▼
[■■·■■] validation_score ■■ → dedup_key ■■■■ → DB ■■
      ■
      ▼
[■■] Slack block-kit payload (mock)
      ■
      ▼
[■■] Streamlit ■■■■ (5■) + FAISS ■■■ ■■
```

■■■ ■■ ■■ Trade-off

■■	■■	■■	■■ ■■
STT	faster-whisper (local)	Whisper API, Clova	■■·■■■■ ■■, large-v3 ■■■ ■■ ■■
■■■■	pyannote 3.1	NeMo, resemblyzer	HuggingFace ■■ ■■ ■■, ■■■ ■■■
LLM ■■	Gemini 2.5 Flash Lite	GPT-4o, Claude	■■ ■■ ■ ■■■ ■■■ ■■ ■■ ■■
DB	SQLite (■■) / PostgreSQL (■■)	DuckDB, MySQL	SQLite■ ■■ ■■ ■■ ■■, Postgres■ ■■ ■■ ■■
■■■■	Streamlit	React, Metabase	Python ■■ ■■, ■■■ ■■ ■■ ■■
■■■	sentence-transformers (local)	Gemini text-embedding	API ■ ■■ ■■ ■■ ■■, 384■■ FAISS IndexFlatIP

---

3. ■■■ ■■■ ■■ ■■

■■■ ■■

```
meetings ■■■ stt_runs
         ■■ participants
         ■■ utterances ■■■■ action_item_sources
         ■■ chunks          (utterance ↔ action ■■■)
         ■■ action_items ■■ action_item_events
                         ■■ slack_payloads
```

- `utterances`■ `action_items`■ ■■■ ■■: STT ■■(■■)■ LLM ■■ ■■(■■)■ ■■■ ■■■ ■■. STT■ ■■■■■ ■■ ■■■ ■■■ ■ ■■.
- `action_item_sources` ■■ ■■■: ■■ ■■■ ■■ ■■■ ■■■■ ■■. ■■■■■■ ■■■ ■■■■ "■■ ■■" ■■■ ■■.
- `action_item_events`: ■■ ■■ ■■(open→in_progress→done)■ ■■ ■■■■ append-only ■■. ■■ ■■■ ■■ ■■ ■■.

■■■■ ■■

`action_items`■ `assignee_normalized`, `campaign_context`, `advertiser_context`■ ■■ ■■■■. ■■■■■ ■■■■ ■■ ■■■■ JOIN■ ■■■ Streamlit ■■ ■■■ ■■■■. ■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■ ■■■■■.

■■ ■■ ■■ ■■

`action_item_id`

```
hash(meeting_id + chunk_id + sequence_no)
```

■■ ■■·■■·■■■■ ■■ ■■ ID → ■■■■■■ ■■■■■ ■■ ■■ ■■(■■■).

`dedup_key`

```
hash(meeting_id + assignee_normalized + category + due_date + normalized_task_signature)
```

■■ ■■ ■■■ LLM■ ■■ ■■■■ ■ ■■■■ ■■ ■■■■ ■■■ ■■■■■ ■■. `action_item_id`(■■ ■■)■ ■■ ■■.

■■■ 3-■■■

■■	■■	■■ ■■
`llm_confidence`	LLM ■■ ■■	■■■■ ■ ■■ ■■ ■■
`validation_score`	■■ ■■ ■■	■■■·■■·■■■■ ■■, ■■ ■■ ■ 5■ ■■
`final_confidence`	■■ ■■■	`min(llm, validation)`

`final_confidence < 0.7` ■■ `risk_flags` ■■ ■ `review_required = true`■ ■■■ ■■■■■ ■■ ■■.

---

4. Before / After [redacted] (100[redacted])

[redacted]

- [redacted]: [redacted] 100[redacted]
- [redacted]: [redacted] 2[redacted]
- [redacted]: [redacted] 45[redacted]
- [redacted]: ~20% ([redacted])
- [redacted]: [redacted] ([redacted])

[redacted]

[redacted]	Before	After	[redacted]
[redacted]	45[redacted]/[redacted]	5[redacted]/[redacted]	[redacted], [redacted]
[redacted] (100[redacted])	—	**6,667[redacted] (111[redacted])**	(45-5)[redacted] × 2[redacted] × 100[redacted]
[redacted]	—	**5,800[redacted]**	111[redacted] × 52[redacted]
[redacted]	~20%	~5%	`review_required` [redacted]
[redacted]	~30%	~8%	`assignee_missing` [redacted]
[redacted]	[redacted] ([redacted])	[redacted]	[redacted] [redacted]·[redacted]

[redacted]

- STT [redacted].
- [redacted] 4[redacted].

---

5. [redacted] + [redacted]

[redacted] A — LLM [redacted] ([redacted]·[redacted])

[redacted]: [redacted] 5[redacted] Gemini[redacted]·[redacted].

[redacted]: `validation_score`[redacted] `llm_confidence`[redacted] 0.2 [redacted] `review_required = true` [redacted]. [redacted] "LLM[redacted]" [redacted].

[redacted]: 1. [redacted]. 2. [redacted] few-shot [redacted]. 3. Gemini API [redacted] Mock Extractor [redacted].

[redacted] B — STT [redacted] ([redacted]·[redacted])

[redacted]: [redacted], [redacted], [redacted]([redacted])[redacted] STT[redacted]. pyannote[redacted] 2[redacted].

[redacted]: STT [redacted] utterance [redacted]. diarization[redacted] "SPEAKER_00" [redacted] fallback [redacted] `stt_runs`[redacted].

[redacted]: 1. [redacted] ([redacted]). 2. STT [redacted] → [redacted]. 3. [redacted](ROAS, CPM, A/B [redacted])[redacted] `glossary.py` [redacted], [redacted].

[redacted] C — Gemini API [redacted]

[redacted]: [redacted](15 RPM) [redacted].

[redacted]: `extraction_runs` [redacted] `error_message` [redacted] API [redacted].

[redacted]: 1. `LLM_PROVIDER=mock` [redacted] Mock [redacted], [redacted]. 2. [redacted](retry_count [redacted]). 3. [redacted] LLM(Ollama + Llama-3)[redacted].

[redacted] 4[redacted]·[redacted]

[redacted]	[redacted]	KPI
1[redacted]	[redacted] 3[redacted]	[redacted], STT [redacted]
2[redacted]	[redacted]	Precision ≥ 0.75, `review_required` [redacted] ≤ 30%
3[redacted]	[redacted]	[redacted], [redacted]

4[redacted]	[redacted]·[redacted]	[redacted], [redacted]